

Asymmetrical effects of rewards and punishments: The case of social loafing

Jennifer M. George*

Department of Management, Texas A&M University, College Station, Texas 77843-4221, USA

An assumption in the social loafing literature is that contingent rewards and contingent punishments have symmetrical effects on the reduction of social loafing in groups. Literature is reviewed from several areas of psychology and organizational behaviour which suggests that rewards and punishments do not have symmetrical effects on individuals and their behaviours. Based upon this review, hypotheses are developed concerning the relationship between supervisor reward and punishment behaviour and social loafing in groups. These hypotheses were tested among a sample of 448 salespeople working in groups ranging in size from four to 10 members. As hypothesized, supervisor contingent reward was negatively associated, and supervisor non-contingent punishment was positively associated with social loafing in ongoing groups. Also as expected, neither supervisor contingent punishment nor non-contingent reward was significantly associated with social loafing. The implications of these results and directions for future research are discussed.

Actual group productivity often falls short of potential productivity (Steiner, 1972). One explanation for this is that individuals exert less effort when they work in groups as opposed to working individually. This phenomenon was studied over 100 years ago by a French agricultural engineer named Ringelmann (Kravitz & Martin, 1986) and has since been called social loafing (Latané, Williams & Harkins, 1979). Latané *et al.* (1979, p. 823) define social loafing as 'a decrease in individual effort due to the social presence of other persons'. Social loafing has been demonstrated to occur on a wide variety of tasks in laboratory settings (e.g. Brickner, Harkins & Ostrom, 1986; Earley, 1989; Jackson & Williams, 1985; Kerr & Bruun, 1981; Petty, Harkins & Williams, 1980; Zaccaro, 1984). Recently, George (1992) demonstrated that some of the results of laboratory studies of social loafing appear to generalize to workers in ongoing groups in organizational settings.

One cause of social loafing is the fact that individual effort or contributions are often perceived to be unidentifiable when work is performed in groups. When individuals in a group think that their efforts or contributions will be identifiable to others and there is the potential for evaluation of these efforts, social loafing is effectively eliminated (e.g. Harkins & Jackson, 1985; Kerr & Bruun, 1981; Williams, Harkins & Latané, 1981).

Why should identifiability be a deterrent to social loafing? The principal explanation for this finding is that when individual contributions are unidentifiable, the perceived

*Requests for reprints.

linkage between individual effort and rewards and punishments is low, resulting in decreased motivation (Jones, 1984). Conversely, when individual effort is perceived to be identifiable, individuals are accountable for their own behaviour and hence can expect resultant positive or negative consequences. As Latané *et al.* (1979, p. 830) indicate, 'since individual scores are unidentifiable when groups perform together, people can receive neither precise credit nor appropriate blame for their performance. . . . Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard'. Hence, one reason individuals engage in social loafing is because they think that their individual efforts will go unrewarded and/or a lack of effort will not be punished (Jones, 1984; Latané *et al.*, 1979). A key question which has heretofore not been addressed is whether contingent rewards and punishments have parallel effects on social loafing. In this regard, converging evidence from several areas in psychology and organizational behaviour suggests that rewards and punishments do not have symmetrical effects on individuals and their behaviours. This suggests that contingent rewards and contingent punishments may not be equally efficacious in curtailing the occurrence of social loafing in organizations.

This paper reviews work in several areas of psychology and organizational behaviour which suggests that rewards and punishments have asymmetrical effects on individuals and their behaviour. Based upon this review, hypotheses are developed concerning the relationship between contingent rewards and contingent punishments and social loafing. Additionally, the effects of non-contingent rewards and punishments on social loafing also are explored.

Asymmetrical effects of rewards and punishments

Evidence suggesting that rewards and punishments have asymmetrical effects can be found in a variety of areas in psychology and organizational behaviour. Below, theory and research from three such areas is briefly reviewed.

Perhaps the most basic argument in support of the asymmetrical effects of rewards and punishments comes from work on brain structure. For example, Gray's theory of personality and emotion (Gray, 1971, 1981, 1987) links rewards and punishments to different types of brain activity or neuronal systems. More specifically, Gray (1981) posits that the behavioural inhibition system influences responses when signals of punishments are present and the behavioural activation system regulates responses when signals of rewards are present (Larsen & Katelaar, 1991). These two physiological systems are independent of each other (Gray, 1981). Gray's theory and research in support of it, therefore, suggest that signals of reward and punishment are processed in different systems in the brain. Hence, rewards and punishment have differential effects on behaviours, in part, because they are responded to by different physiological systems.

Any discussion of rewards and punishments brings to mind the operant conditioning paradigm, a second topical area dealing with the asymmetrical effects of rewards and punishments. From this perspective, rewards are seen as increasing the probability of a behaviour and punishments are viewed as decreasing the probability of a behaviour. However, it has long been recognized that the effects of punishment are distinct from the effects of rewards in that they do not have parallel (or opposite) influences on behaviour.

For example, Skinner (1953, p. 183) noted that 'in the long run, punishment, unlike reinforcement, works to the disadvantage of both the punished organism and the punishing agency. The aversive stimuli which are needed generate emotions, including predispositions to escape or retaliate, and disabling anxieties . . . the suspicion has also arisen that punishment does not in fact do what it is supposed to do'. Skinner goes on to discuss a variety of unintended negative consequences of punishment. This point also has been acknowledged in the organizational literature on operant conditioning. For example, Fedor & Ferris (1981) discuss several problems with the use of punishment in organizations including the potential for retaliatory behaviours. More generally, organizational behaviour modification discourages the use of punishment (Fedor & Ferris, 1981).

Finally, research on the effects of leader reward and punishment behaviour suggests that rewards and punishments do not have symmetrical effects on behaviour. For example, Podsakoff, Todor, Grover & Huber (1984) review results which suggest that contingent rewards consistently have positive effects on performance whereas the effects of contingent punishment are much less clear cut (e.g. Cherrington, Reitz & Scott, 1971; Greene, 1973, 1979; Podsakoff, Todor & Skov, 1982; Sims, 1977; Sims & Szilagyi, 1975). Consistent with this reasoning, Sims (1980, p. 134) concluded in his review that 'the relationship between reward behavior and subordinate performance is much stronger than the relationship between punitive behavior and performance'. More recently, Podsakoff *et al.* (1984) found that leader contingent reward behaviour was positively associated with subordinate performance while contingent punishment behaviour was found to be unrelated to performance. Additionally, Podsakoff *et al.* (1984) found that non-contingent reward behaviour was unrelated to performance whereas leader non-contingent punishment behaviour was negatively associated with subordinate performance.

In summary, theory and research from several areas of psychology and organizational behaviour suggest that rewards and punishments do not have symmetrical effects on individuals and their behaviours. Hence, it stands to reason that contingent rewards and punishments would not be equally efficacious in terms of reducing social loafing in organizations, contrary to a common assumption in this literature (e.g. Latané *et al.*, 1979).

Effects of contingent rewards and contingent punishments on social loafing

When a supervisor or leader contingently rewards a subordinate, he or she conveys several pieces of information to that subordinate. Principal among these is the fact that the subordinate's behaviour or actions have been noticed by the supervisor and are considered to be important or valued. Contingent rewards also give subordinates information or feedback concerning their competence and effectiveness. Hence, when supervisors contingently reward subordinates, even if work is performed in groups, subordinates are likely to perceive that their own efforts are identifiable (i.e. they would have to be to be contingently rewarded for them) and that they will receive rewards for them. Social loafing should be less likely under these circumstances since individuals will be more confident that their efforts will be appropriately rewarded (Shepperd, 1993). Both the operant conditioning paradigm and the social loafing literature are supportive of this expectation (e.g. Latané *et al.*, 1979). Thus:

Hypothesis 1: The extent to which supervisors contingently reward subordinates based upon individual performance is negatively associated with social loafing.

What about the effects of contingent punishment on social loafing? While intuitively we might expect them to be similar in that when a supervisor contingently punishes a subordinate for substandard performance, social loafing should be less prevalent, we know from the literature discussed above that punishment appears to have phenomenologically distinct effects from rewards. For example, the research reviewed above on brain structure, operant conditioning and leader behaviour all suggest that rewards and punishments have asymmetrical effects.

Deci's cognitive evaluation theory (e.g. Deci, 1971; Deci, Connell & Ryan, 1989; Deci & Ryan, 1980) provides a useful framework from which to explore these effects in the case of social loafing. Two premises of this theory, which are also echoed in other psychological approaches to human nature, are that people have a need for self-determination or like to believe that they have control over their own actions, and that people also like to feel that they are competent. Contingent punishment is used precisely when one party has control or power over the other (Skinner, 1953) and, in the case of the supervisor-subordinate relationship, is likely to drive home to the subordinate the fact that his or her supervisor has control. In the case of social loafing, this puts the subordinate in a bit of a quandary. On the one hand, contingent punishment sends the signal that social loafing will be noticed and negative consequences may ensue. On the other hand, the subordinate's needs for self-determination may be thwarted if he or she succumbs to the control of the supervisor's punishing behaviour. For example, in response to such a situation, one way to assert one's self-determination would be to engage in social loafing. Consistent with this reasoning is Miles & Greenberg's (1993) unexpected finding in their study of social loafing that higher levels of punishment threats lead to lower levels of individual performance among swimmers. Miles & Greenberg posit that such threats may be seen as challenges to individual freedom and one way to re-establish such freedom is through lower performance (p. 260).

The idea that contingent punishment generates competing motives for and against social loafing is also somewhat consistent with Shepperd's (1993) recent motivational analysis of social loafing. Building from Shepperd's (1993) approach, contingent punishment may discourage social loafing to the extent that it increases the value of individual effort expenditure (e.g. as a means to avoid punishment). On the other hand, contingent punishment may encourage social loafing to the extent that it increases the cost of effort expenditure (e.g. it is unpleasant to work hard for someone who has recently punished you). Given these conflicting tendencies towards and against social loafing engendered by contingent punishment and the fact that the supervisor can not realistically punish every single instance of social loafing (Skinner, 1953), it is likely that contingent punishment will show no appreciable relationship to social loafing in a positive or negative direction. In particular, contingent punishment behaviour is unlikely to reduce social loafing, contrary to an assumption in the social loafing literature (e.g. Latané *et al.*, 1979). Thus:

Hypothesis 2: The extent to which supervisors contingently punish subordinates based upon individual performance is unrelated to social loafing.

At this point, it should be noted that contingent reward behaviour is not expected to

threaten workers' feelings of self-determination (as in the case of contingent punishment described above) due to the fact that the contingent rewards investigated served an informational function and provided workers with feedback on their actions and capabilities (Deci & Ryan, 1980). In providing information concerning one's effectiveness and/or capabilities, rewards do not threaten self-determination and may enhance feelings of competence leading to increased motivation (Deci & Ryan, 1980). As Deci & Ryan (1980, p. 63) indicate, 'positive competence feedback should always increase intrinsic motivation'. While one could argue that contingent punishment also provides subordinates with information, the fact that this information is very likely to be negative will not enhance feelings of competence. Moreover, the controlling nature of the punishment is likely to be most salient.

Effects of non-contingent rewards and non-contingent punishments on social loafing

Podsakoff *et al.* (1984) emphasized the need to consider the effects of non-contingent reward and punishment behaviour on subordinate responses in addition to the effects of contingent reward and punishment behaviour. In the case of non-contingent reward behaviour, since rewards are administered irrespective of performance levels, social loafing should be unaffected. Alternatively, it could be argued that non-contingent reward behaviour may actually encourage social loafing since workers may perceive that they will receive rewards regardless of their efforts and thus can get away with social loafing. Workers also may feel that their efforts are unnecessary when they are non-contingently rewarded, encouraging social loafing (Shepperd, 1993).

However, a countervailing force is likely to offset the incentive to engage in social loafing under conditions of non-contingent reward. That is, theories of social exchange suggest that people seek to reciprocate with others from whom they receive benefits or rewards (Blau, 1964). Hence, in the case of social loafing, whereas non-contingent reward behaviour may encourage social loafing to the extent that it signals that one can get away with it, it also may discourage social loafing to the extent that workers feel the need to reciprocate with their benefactors (i.e. their rewarding supervisors), one of the most readily available means of reciprocating being effort on the job. Thus, while from a strict operant conditioning paradigm non-contingent reward should have no effect on social loafing since it is, by definition, not conditional upon individual contributions, any incentive to social loafing engendered by such reward is likely to be offset by desires for reciprocity in social exchange (Blau, 1964). Thus:

Hypothesis 3: The extent to which supervisors non-contingently reward individual subordinates is unrelated to social loafing.

Non-contingent punishment behaviour occurs when a supervisor punishes a subordinate for no apparent reason. While such behaviour is likely to engender negative emotional reactions on the part of the subordinate, it also is likely to have behavioural implications (e.g. Podsakoff *et al.*, 1984). For example, receiving non-contingent punishment may cause workers to desire to retaliate against the supervisor or the organization. From an equity theory perspective, receiving non-contingent punishment is likely to result in perceptions of unfairness and the resultant desire to restore equity (Adams,

1963). In such a situation, one way to restore equity would be to lower one's efforts by engaging in social loafing. Thus:

Hypothesis 4: The extent to which supervisors non-contingently punish individual subordinates is positively associated with social loafing.

The results of a study designed to test these hypotheses are described and discussed below.

Method

Data for this study came from a larger study conducted in eight branches of a major retailer in the United States. The data used in this study have not been previously published in any form. The sample consisted of salespeople who worked in primary work-groups ranging in size from four to 10 members. A primary work-group was defined as a group of salespeople who worked together, shared group responsibilities, worked together towards the attainment of group goals, and whose members and management at the retailer were viewed as being in the same work-group. The salespeople always worked in their primary work-groups. The human resources department, in conjunction with branch management, identified each of the groups and their members who were included in the study. All primary work-groups in the eight branches for which there were no ambiguities regarding group membership were included in the study.

The salespeople were given questionnaires at work with a postage-paid return envelope to return their completed questionnaires directly to the researcher. The questionnaires included measures of the extent to which a salesperson's supervisor engaged in contingent reward behaviour, contingent punishment behaviour, non-contingent reward behaviour and non-contingent punishment behaviour with regard to the salesperson's own on-the-job behaviours. Participation was voluntary and complete confidentiality was guaranteed. A total of 579 questionnaires were distributed and 448 completed questionnaires were returned for a 77 per cent response rate. Approximately 85 per cent of the respondents were female. The average age of the respondents was 41 years and 53 per cent of the sample reported having attended a college or technical school.

While the salespeople worked in primary work-groups, discussions with top, middle and lower level management at the retailer and the supervisors themselves indicated that the supervisors of the salespeople were aware of individual effort and performance levels. Indeed, part of the supervisors' responsibilities was the evaluation of the contributions of individual salespersons to the group and organization. Discussions with the supervisors indicated that social loafing was a very real phenomenon in the groups they managed and confirmed that they had ample opportunity to observe it. The supervisors of the salespeople received a rating form for each of their subordinates who was included in the study. The rating forms included a measure of the extent to which each of the salespersons engaged in social loafing. The supervisors were given the rating forms at work with a postage-paid envelope to return the completed forms directly to the researcher. Participation was voluntary and respondents were guaranteed complete confidentiality. Sixty-four supervisors received rating forms and 53 returned completed forms for a response rate of 83 per cent. Because of missing data (e.g. receipt of a subordinate's questionnaire but no corresponding rating form from the subordinate's supervisor and vice versa), the sample size for analyses ranged from 345 to 448.

Measures

Contingent reward behaviour, contingent punishment behaviour, non-contingent reward behaviour and non-contingent punishment behaviour. The extent to which the supervisors of the salespeople engaged in these behaviours was measured with scales described by Podsakoff *et al.* (1984). The Contingent Reward Behaviour scale contained 10 items, the Contingent Punishment Behaviour scale contained five items, and the Non-contingent Reward and Punishment scales each contained four items. Information on the development and psychometric properties of these scales is provided by Podsakoff *et al.* (1984). Sample items from the Contingent Reward Behaviour scale are: 'My supervisor commends me when I do a better than average job', and 'My supervisor gives me special recognition when my work performance is especially good'. Responses were made on a seven-point scale ranging from 'strongly disagree' to 'strongly agree'.

Social loafing. Social loafing was measured by the supervisors of the salespeople with the 10-item Social Loafing scale developed by George (1992). This scale measures the extent to which a worker puts forth low effort on the job when others are present to do the work. Information on the development and psychometric properties of the Social Loafing scale is provided by George (1992). A sample item is: 'Puts forth less effort on the job when other salespeople are around to do the work'. The supervisors were instructed to rate how characteristic of the salesperson being evaluated each of the items was on a five-point scale ranging from 'not at all characteristic' to 'very characteristic'. The complete Social Loafing scale is provided in the Appendix.

Results

Means, standard deviations, internal consistency reliabilities and intercorrelations among the study variables are presented in Table 1. All of the internal consistency reliabilities for the scales used were acceptable (Nunnally, 1978). Hypotheses 1 and 2 concerned the relationships between contingent reward behaviour and contingent punishment behaviour and social loafing. In support of hypothesis 1, contingent reward behaviour was significantly and negatively associated with social loafing ($r = -.26$; $p \leq .001$). Consistent with hypothesis 2, contingent punishment behaviour showed no appreciable relationship to social loafing ($r = -.07$; n.s.).

Hypotheses 3 and 4 pertained to the associations between non-contingent reward behaviour and non-contingent punishment behaviour and social loafing. Consistent with hypothesis 3, non-contingent reward behaviour was found to be unrelated to social loafing ($r = .02$; n.s.). In support of hypothesis 4 non-contingent punishment behaviour was significantly and positively related to social loafing ($r = .25$; $p \leq .001$).

Hypotheses 1 through 4 can also be tested using multiple regression. Table 2 contains the results of regressing social loafing simultaneously on contingent reward behaviour, contingent punishment behaviour, non-contingent reward behaviour, and non-contingent punishment behaviour. In support of hypotheses 1 and 4, the beta weights for contingent reward behaviour ($\beta = -.15$; $p \leq .05$) and non-contingent punishment behaviour ($\beta = .18$; $p \leq .01$) were each statistically significant and in the expected

Table 1. Summary statistics

Variables	Means	SD	Correlations ^a				
			1	2	3	4	5
1. Contingent reward behaviour	44.33	13.75	(.95)	.15***	.14***	-.54***	-.26***
2. Contingent punishment behaviour	25.38	5.35		(.86)	-.29***	.12**	-.07
3. Non-contingent reward behaviour	10.83	4.47			(.77)	.02	.02
4. Non-contingent punishment behaviour	10.19	5.01				(.88)	.25***
5. Social loafing	17.34	7.83					(.93)

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$.

^a Coefficient alphas are in parentheses on the diagonal.

Table 2. Results of regressing social loafing on leader behaviours

Independent variables	Beta
Contingent reward behaviour	-.15*
Contingent punishment behaviour	-.07
Non-contingent reward behaviour	.01
Non-contingent punishment behaviour	.18**
R = .30***	
R ² = .09***	

* $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$.

directions. Consistent with hypotheses 2 and 3, contingent punishment behaviour and non-contingent reward behaviour showed no appreciable relationship to social loafing.

Finally, results highly consistent with those reported in Table 2 were found controlling for branch. That is, in a regression equation in which branch membership was represented by seven dummy-coded variables (Cohen & Cohen, 1983), the beta weights for contingent reward behaviour ($\beta = -.16$; $p \leq .01$) and non-contingent punishment behaviour ($\beta = .18$; $p \leq .01$) were each statistically significant and in the hypothesized direction while the betas for contingent punishment behaviour and non-contingent reward behaviour were non-significant. This supplemental analysis was performed in case there was some difference across branches which was, in fact, partially responsible for the results obtained. Such a difference was not expected and the results were consistent with this expectation. Once again, these results support hypotheses 1–4.

Discussion

Consistent with a dominant assumption in the social loafing literature (e.g. Jones, 1984; Latané *et al.*, 1979), a supervisor's contingent reward behaviour was found to be negatively associated with social loafing in ongoing groups. Hence, a supervisor who positively reinforces desirable behaviour appears to help curtail the occurrence of social loafing in groups. Importantly, for supervisor contingent reward behaviour to be effective, it is necessary that workers perceive that rewards are, in fact, contingently administered.

Since social loafing is an undesirable behaviour, intuitively one might expect that contingent punishment would help to curtail social loafing. Indeed, the social loafing literature assumes that contingent rewards and punishments will have symmetrical effects on social loafing (e.g. Harkins, Latané & Williams, 1980; Latané *et al.*, 1979). However, the effects of punishment are not necessarily simple or straightforward as suggested by the literature reviewed earlier. Rewards and punishments do not have symmetrical effects on behaviours and punishment appears to engender unintended negative consequences. In the case of social loafing, a supervisor's contingent punishment behaviour does not appear to be a deterrent, as hypothesized. Hence, while a supervisor's first reaction to substandard performance may be to reprimand a worker, in the long run such behaviours are not likely to be as effective as noticing and reinforcing desirable behaviours.

In this study, while the work was performed in groups, supervisors were aware of individual effort and contributions. However, in other contexts it is likely that supervisors

may not be able to determine individual effort or contributions to group performance. This may be the case, for example, when a group of workers is physically separated from the supervisor or when the nature of the group work necessitates a high degree of reciprocal interdependence among group members (Jones, 1984; Thompson, 1967). Under those circumstances, supervisors cannot engage in contingent reward behaviour at the level of the individual worker since individual contributions cannot be isolated. While supervisors can certainly administer group-based rewards contingent upon group performance, social loafing may still be a problem since rewards are administered to the group as a whole and not to individuals (Jones, 1984). At the individual level, supervisors may be tempted to use non-contingent rewards; however, the results of this study suggest that this may have no appreciable effect on social loafing. Finally, a supervisor's non-contingent punishment behaviour may increase the incidence of social loafing.

Additionally, it should be noted that on some kinds of jobs workers receive direct feedback in the course of performing work tasks. On these kinds of jobs, the informational value of rewards from supervisors is lessened because workers already know how they are doing. Under these conditions, contingent rewards may have somewhat different effects on the incidence of social loafing.

Given what seems to be an increasing reliance on groups and teams in organizations (or at least the espousing of the benefits of group work), it is surprising that very little research has been conducted on social loafing in ongoing groups. In fact, with few exceptions (e.g. George, 1992; Hardy & Latané, 1988; Williams, Nida, Baca & Latané, 1989), social loafing has been studied exclusively in laboratory settings. Moreover, studies of social loafing in ongoing groups have not tended to focus on work-groups *per se* (e.g. Hardy & Latané, 1988; Williams *et al.*, 1989). If group work is becoming more popular, then social loafing is an important issue organizations will have to face. As mentioned earlier, the ubiquity of social loafing has been demonstrated for many different types of tasks in laboratory settings (e.g. Earley, 1989; Harkins *et al.*, 1980). Further research is needed which focuses on the occurrence of social loafing in groups in organizations and ways to reduce it.

Social loafing is not just a problem in terms of the lost productivity of the individuals who engage in social loafing. Rather, if some group members engage in social loafing, others may reduce their own effort levels so as not to be 'suckers' (Kerr, 1983). For example, research conducted by Jackson & Harkins (1985) indicates that when group members expect others to engage in social loafing they may reduce their own efforts to approximate the effort levels of the social loafers. Hence, social loafing by one or a few members of a group may cause others to lower their own efforts as well.

Once again, while results of the current study suggest that supervisor contingent reward behaviour is an effective means of reducing social loafing, when individual contributions cannot be identified, this is not an option. Future research is needed to uncover other ways to help reduce the occurrence of social loafing under these circumstances. For example, if group members themselves are able to assess individual contributions, then group-administered contingent rewards (to individuals based upon their individual contributions) may be one option to consider. A precondition necessary for such an option to result in outcomes consistent with organizational goals is that group goals are congruent with the organization's goals. The efficacy of this and other ways to help reduce the occurrence of social loafing in such contexts is an empirical question in need of future research.

This study is not without limitations. For example, due to the non-experimental nature of the data, the direction of causality cannot be determined unambiguously. For example, rather than supervisor reward and punishment behaviour having an effect on the incidence of social loafing, it may be that social loafing is influencing supervisor behaviour. However, an examination of the pattern of correlations observed partially mitigates this concern. That is, if social loafing was determining supervisor behaviour, then one would expect to find a positive association between social loafing and contingent punishment behaviour such that when individuals engaged in social loafing their supervisors were more likely to punish them contingently. However, consistent with hypothesis 2 and the theoretical reasoning behind it, contingent punishment was found to be unrelated to social loafing. Another limitation of the study is the fact that two of the hypotheses (hypotheses 2 and 3) are null hypotheses or predict the absence of a significant relationship. These and other potential limitations notwithstanding, it is hoped that this study prompts other theorists and researchers to explore social loafing in ongoing work-groups and ways to eliminate it or at least reduce its occurrence.

References

- Adams, J. S. (1963). Toward an understanding of inequity. *Journal of Abnormal and Social Psychology*, 67, 422–436.
- Blau, P. M. (1964). *Exchange and Power in Social Life*. New York: Wiley.
- Brickner, M. A., Harkins, S. G. & Ostrom, T. M. (1986). Effects of personal involvement: Thought-provoking implications for social loafing. *Journal of Personality and Social Psychology*, 51, 763–769.
- Cherrington, D. J., Reitz, H. J. & Scott, W. E. (1971). Effects of contingent and noncontingent rewards on the relationship between satisfaction and performance. *Journal of Applied Psychology*, 55, 531–536.
- Cohen, J. & Cohen, P. (1983). *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*. Hillsdale, NJ: Erlbaum.
- Davis, J. H. (1969). *Group Performance*. Reading, MA: Addison-Wesley.
- Deci, E. L. (1971). Effects of externally mediated rewards on intrinsic motivation. *Journal of Applied Psychology*, 18, 105–115.
- Deci, E. L., Connell, J. P. & Ryan, R. M. (1989). Self-determination in a work organization. *Journal of Applied Psychology*, 74, 580–590.
- Deci, E. L. & Ryan, R. M. (1980). The empirical exploration of intrinsic motivational processes. *Advances in Experimental Social Psychology*, 13, 39–80.
- Earley, P. C. (1989). Social loafing and collectivism: A comparison of the United States and the People's Republic of China. *Administrative Science Quarterly*, 34, 565–581.
- Fedor, D. B. & Ferris, G. R. (1981). Integrating OB MOD with cognitive approaches to motivation. *Academy of Management Review*, 6, 115–125.
- George, J. M. (1992). Extrinsic and intrinsic origins of perceived social loafing in organizations. *Academy of Management Journal*, 35, 191–202.
- Gray, J. A. (1971). The psychophysiological basis of introversion–extraversion. *Behaviour Research and Therapy*, 8, 249–266.
- Gray, J. A. (1981). A critique of Eysenck's theory of personality. In H. J. Eysenck (Ed.), *A Model for Personality*, pp. 246–276. New York: Springer.
- Gray, J. A. (1987). Perspectives on anxiety and impulsivity: A commentary. *Journal of Research in Personality*, 21, 493–509.
- Greene, C. N. (1973). Causal connections among managers' merit pay, job satisfaction, and performance. *Journal of Applied Psychology*, 58, 95–100.
- Greene, C. N. (1979). Questions of causation in the path-goal theory of leadership. *Academy of Management Journal*, 22, 22–41.
- Hardy, C. J. & Latané, B. (1988). Social loafing in cheerleaders: Effects of team membership and competition. *Journal of Sport & Exercise Psychology*, 10, 109–114.

- Harkins, S. & Jackson, J. (1985). The role of evaluation in eliminating social loafing. *Personality and Social Psychology Bulletin*, 11, 457–465.
- Harkins, S. G., Latané, B. S. & Williams, K. (1980). Social loafing: Allocating effort or taking it easy. *Journal of Experimental Social Psychology*, 16, 457–465.
- Jackson, J. M. & Harkins, S. G. (1985). Equity in effort: An explanation of the social loafing effect. *Journal of Personality and Social Psychology*, 49, 1199–1206.
- Jackson, J. M. & Williams, K. D. (1985). Social loafing on difficult tasks: Working collectively can improve performance. *Journal of Personality and Social Psychology*, 49, 937–942.
- Jones, G. R. (1984). Task visibility, free riding and shirking: Explaining the effect of structure and technology on employee behavior. *Academy of Management Review*, 9, 684–695.
- Kerr, N. L. (1983). Motivation losses in small groups: A social dilemma analysis. *Journal of Personality and Social Psychology*, 45, 819–828.
- Kerr, N. L. & Bruun, S. E. (1981). Ringelmann revisited: Alternative explanations for the social loafing effect. *Personality and Social Psychology Bulletin*, 7, 224–231.
- Kravitz, D. A. & Martin, B. (1986). Ringelmann rediscovered: The original article. *Journal of Personality and Social Psychology*, 50, 936–941.
- Larsen, R. J. & Katelaar, T. (1991). Personality and susceptibility to positive and negative emotional states. *Journal of Personality and Social Psychology*, 61, 132–140.
- Latané, B., Williams, K. D. & Harkins, S. (1979). Many hands make light the work: The causes and consequences of social loafing. *Journal of Personality and Social Psychology*, 37, 822–832.
- Miles, J. A. & Greenberg, J. (1993). Using punishment threats to attenuate social loafing effects among swimmers. *Organizational Behavior and Human Decision Processes*, 56, 246–265.
- Nunnally, J. C. (1978). *Psychometric Theory*. New York: McGraw-Hill.
- Petty, R. E., Harkins, S. G. & Williams, K. D. (1980). The effects of group diffusion of cognitive effort on attitudes: An information processing view. *Journal of Personality and Social Psychology*, 3, 579–582.
- Podsakoff, P. M., Todor, W. D., Grover, R. A. & Huber, V. L. (1984). Situational moderators of leader reward and punishment behaviors: Fact or fiction? *Organizational Behavior and Human Performance*, 34, 21–63.
- Podsakoff, P. M., Todor, W. D. & Skov, R. (1982). Effects of leader contingent and noncontingent reward and punishment behaviors on subordinate performance and satisfaction. *Academy of Management Journal*, 25, 810–821.
- Shepperd, J. A. (1993). Productivity loss in performance groups: A motivational analysis. *Psychological Bulletin*, 113, 67–81.
- Sims, H. P. Jr (1977). The leader as a manager of reinforcement contingencies: An empirical example and a model. In J. G. Hunt & L. L. Larson (Eds), *Leadership: The Cutting Edge*. Carbondale, IL: Southern Illinois University Press.
- Sims, H. P. Jr (1980). Further thoughts on punishment in organizations. *Academy of Management Review*, 5, 133–138.
- Sims, H. P. & Szilagyi, A. D. (1975). Leader reward behavior and subordinate satisfaction and performance. *Organizational Behavior and Human Performance*, 14, 426–438.
- Skinner, B. F. (1953). *Science and Human Behavior*. New York: Macmillan.
- Steiner, I. (1972). *Group Processes and Productivity*. San Diego, CA: Academic Press.
- Thompson, J. D. (1967). *Organizations in Action*. New York: McGraw-Hill.
- Williams, K., Harkins, S. & Latané, B. (1981). Identifiability as a deterrent to social loafing: Two cheering experiments. *Journal of Personality and Social Psychology*, 40, 303–311.
- Williams, K. D., Nida, S. A., Baca, L. D. & Latané, B. (1989). Social loafing and swimming: Effects of identifiability of individual and relay performance of intercollegiate swimmers. *Basic and Applied Social Psychology*, 10, 73–81.
- Zaccaro, S. J. (1984). Social loafing: The role of task attractiveness. *Personality and Social Psychology Bulletin*, 10, 99–106.

Appendix

The Social Loafing scale

1. Defers responsibilities he or she should assume to other salespeople.
2. Puts forth less effort on the job when other salespeople are around to do the work.
3. Does not do his or her share of the work.
4. Spends less time helping customers if other salespeople are present to serve customers.
5. Puts forth less effort than other members of his or her work-group.
6. Avoids performing housekeeping tasks as much as possible.
7. Leaves work for the next shift which he or she should really complete.
8. Is less likely to approach a customer if another salesperson is available to do this.
9. Takes it easy if other salespeople are around to do the work.
10. Defers customer service activities to other salespeople if they are present.

Supervisors are instructed to indicate how characteristic each of the items are of the salesperson they are rating on the following five-point scale:

1	2	3	4	5
Not at all characteristic	Slightly characteristic	Somewhat characteristic	Characteristic	Very characteristic

Responses to the 10 items are summed for an overall score.

Source: George (1992).